第一部分 高等数学

一 函数、极限、连续

一、选择题

1.(D). 题目中函数为偶函数.选(D).

2.(D). 当 $x\to 0$ 时,$\sqrt{\dfrac{1+x}{1-x}}-1=\left(1+\dfrac{2x}{1-x}\right)^{\frac{1}{2}}-1\sim\dfrac{1}{2}\cdot\dfrac{2x}{1-x}\sim x$;

$e^{x}-e^{\ln(1+x)}=e^{\ln(1+x)}\cdot[e^{x-\ln(1+x)}-1]\sim e^{x-\ln(1+x)}-1\sim x-\ln(1+x)\sim\dfrac{1}{2}x^{2}$;

$\cos x+e^{x^{2}}-2=(\cos x-1)+(e^{x^{2}}-1)\sim-\dfrac{1}{2}x^{2}+x^{2}=\dfrac{1}{2}x^{2}$;

$\tan x-\sin x=\tan x(1-\cos x)\sim x\cdot\dfrac{1}{2}x^{2}=\dfrac{1}{2}x^{3}$.选(D).

3.(C). $$\lim_{x\to 0}\frac{\alpha}{x}=\lim_{x\to 0}\frac{\ln(x+\sqrt{1+x^{2}})}{x}=\lim_{x\to 0}\frac{\ln[1+(x+\sqrt{1+x^{2}}-1)]}{x}$$

$$=\lim_{x\to 0}\frac{x+\sqrt{1+x^{2}}-1}{x}=1+\lim_{x\to 0}\frac{\sqrt{1+x^{2}}-1}{x}$$

$$=1+\lim_{x\to 0}\frac{(1+x^{2})^{\frac{1}{2}}-1}{x}=1+\lim_{x\to 0}\frac{\frac{1}{2}x^{2}}{x}=1,$$

则 $\alpha\sim x$.

$$\beta=\frac{\sqrt{1+x}-\sqrt{1+x\cos x}}{\sqrt{1+\sin x}-\sqrt{1+\ln(1+x)}}=\frac{\sqrt{1+\sin x}+\sqrt{1+\ln(1+x)}}{\sqrt{1+x}+\sqrt{1+x\cos x}}\cdot\frac{x(1-\cos x)}{\sin x-\ln(1+x)},$$

$$\lim_{x\to 0}\frac{\beta}{x}=\lim_{x\to 0}\frac{1-\cos x}{\sin x-\ln(1+x)}=\frac{1}{2}\lim_{x\to 0}\frac{x^{2}}{\sin x-\ln(1+x)}=\frac{1}{2}\lim_{x\to 0}\frac{2x}{\cos x-\frac{1}{1+x}}$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{2}{-\sin x+\frac{1}{(1+x)^{2}}}=1,$$

则 $\beta\sim x$,故 $\alpha\sim\beta$.选(C).

4.(C). 当 $x\to 0^{+}$ 时,$\alpha=\ln\cos x=\ln[1+(\cos x-1)]\sim\cos x-1\sim-\dfrac{1}{2}x^{2}$,

$$\beta=(1+x)^{\frac{2}{x}}-e^{2}=e^{2}\left[e^{\frac{2\ln(1+x)}{x}-2}-1\right]\sim 2e^{2}\cdot\frac{\ln(1+x)-x}{x}\sim-e^{2}x,$$

$$\lim_{x\to 0}\frac{\gamma}{x^{3}}=\lim_{x\to 0}\frac{\sqrt{1+x\cos x}-\sqrt{1+\sin x}}{x^{3}}=\lim_{x\to 0}\frac{1}{\sqrt{1+x\cos x}+\sqrt{1+\sin x}}\cdot\frac{x\cos x-\sin x}{x^{3}}$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{x\cos x-\sin x}{x^{3}}=\frac{1}{6}\lim_{x\to 0}\frac{\cos x-x\sin x-\cos x}{x^{2}}=-\frac{1}{6},$$

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则 $\gamma \sim -\frac{1}{6}x^3$. 选(C).

[注] 当 $x \to 0^+$ 时，$\sqrt{1+x\cos x} - \sqrt{1+\sin x} = \frac{x\cos x - \sin x}{\sqrt{1+x\cos x}+\sqrt{1+\sin x}} \sim \frac{1}{2}\cos x(x-\tan x) \sim \frac{1}{2}(x-\tan x)$，根据所学内容，$x$，$\sin x$，$\tan x$，$\arcsin x$，$\arctan x$ 任意两个相减都是 x^3 的同阶无穷小，可以直接比较出 α,β,γ 的阶数，不一定要算出最后的结果.

5.(B). 由 $(x+1)^a = x^a + ax^{a-1} + \cdots$，$(x-1)^a = x^a - ax^{a-1} + \cdots$，得

$$(x+1)^a - (x-1)^a = 2ax^{a-1} + \cdots,$$

则 $\begin{cases} a-1=4, \\ b=\frac{2}{2a}, \end{cases}$ 故 $a=5, b=\frac{1}{5}$. 选(B).

6.(A). $x - \sin x\cos x\cos 2x = x - \frac{1}{2}\sin 2x\cos 2x = \frac{1}{4}\cdot 4x - \frac{1}{4}\sin 4x$.

x，$\sin x$，$\tan x$，$\arcsin x$，$\arctan x$ 任意两个相减都是 x^3 的同阶无穷小，则

$$\lim_{x\to 0}\frac{x-\sin x\cos x\cos 2x}{x^3} = \frac{1}{4}\lim_{x\to 0}\frac{4x-\sin 4x}{x^3} = 16\lim_{x\to 0}\frac{4x-\sin 4x}{(4x)^3}$$

$$\xlongequal{t=4x} 16\lim_{t\to 0}\frac{t-\sin t}{t^3} = \frac{16}{3}\lim_{t\to 0}\frac{1-\cos t}{t^2} = \frac{8}{3},$$

故 $a=\frac{8}{3}, b=3$. 选(A).

7.(A). $\lim\limits_{x\to\infty}(\sqrt[3]{x^6-4x^3+5} - ax^2 - b) = \lim\limits_{x\to\infty}x^2\left[\left(1-\frac{4}{x^3}+\frac{5}{x^6}\right)^{\frac{1}{3}} - a - \frac{b}{x^2}\right] = \lim\limits_{x\to\infty}(1-a)x^2$，

当且仅当 $a=1$ 时，极限存在，则

$$\text{原式} = \lim_{x\to\infty}x^2\left[\left(1-\frac{4}{x^3}+\frac{5}{x^6}\right)^{\frac{1}{3}} - 1 - \frac{b}{x^2}\right] = \lim_{x\to\infty}\frac{\left(1-\frac{4}{x^3}+\frac{5}{x^6}\right)^{\frac{1}{3}} - 1 - \frac{b}{x^2}}{\frac{1}{x^2}}$$

$$\xlongequal{t=\frac{1}{x}} \lim_{t\to 0}\frac{(1-4t^3+5t^6)^{\frac{1}{3}} - 1 - bt^2}{t^2} = \lim_{t\to 0}\frac{\frac{1}{3}(-4t^3+5t^6) - bt^2}{t^2} = -b = 2,$$

故 $b=-2$. 选(A).

8.(D). $\lim\limits_{x\to 1}\frac{2x^2-x-1}{x^2-1} = \lim\limits_{x\to 1}\frac{2x+1}{x+1} = \frac{3}{2}$，$\lim\limits_{x\to 1^-}\frac{2-e^{\frac{1}{x-1}}}{1+e^{\frac{1}{x-1}}} = 2$，$\lim\limits_{x\to 1^+}\frac{2-e^{\frac{1}{x-1}}}{1+e^{\frac{1}{x-1}}} = -1$，则

$$\lim_{x\to 1^-}\frac{2x^2-x-1}{x^2-1}\cdot\frac{2-e^{\frac{1}{x-1}}}{1+e^{\frac{1}{x-1}}} = 3 \neq \lim_{x\to 1^+}\frac{2x^2-x-1}{x^2-1}\cdot\frac{2-e^{\frac{1}{x-1}}}{1+e^{\frac{1}{x-1}}} = -\frac{3}{2}.$$

选(D).

9.(D). 由题意知，$\lim\limits_{x\to 0}f(x) = f(0) = 0$，则

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$$\lim_{x\to 0}[e^{2x}+f(x)]^{\frac{1}{\arcsin x}}=\lim_{x\to 0}\left\{[1+e^{2x}-1+f(x)]^{\frac{1}{e^{2x}-1+f(x)}}\right\}^{\frac{e^{2x}-1+f(x)}{\arcsin x}}=e^{\lim\limits_{x\to 0}\frac{e^{2x}-1+f(x)}{\arcsin x}}$$

$$=e^{\lim\limits_{x\to 0}\frac{e^{2x}-1+f(x)}{x}}=e^{\lim\limits_{x\to 0}\frac{e^{2x}-1}{x}+\lim\limits_{x\to 0}\frac{f(x)}{x}}=e^{4}.$$

选(D).

10. (**A**). $f(0-0)=\lim\limits_{x\to 0^-}f(x)=\lim\limits_{x\to 0^-}\left\{[1+(\cos x-1)]^{\frac{1}{\cos x-1}}\right\}^{\frac{1-\cos x}{x^2}}=e^{\frac{1}{2}}$;

$$f(0+0)=\lim_{x\to 0^+}f(x)=\lim_{x\to 0^+}\left[\left(1+\frac{\arctan x-x}{x}\right)^{\frac{x}{\arctan x-x}}\right]^{b\frac{\arctan x-x}{x^3}}$$

$$=e^{b\lim\limits_{x\to 0^+}\frac{\arctan x-x}{x^3}}=e^{\frac{b}{3}\lim\limits_{x\to 0^+}\frac{\frac{1}{1+x^2}-1}{x^2}}=e^{-\frac{b}{3}},$$

因为 $f(x)$ 在 $x=0$ 处连续,所以 $f(0-0)=f(0)=f(0+0)$,故 $a=e^{\frac{1}{2}}$,$b=-\frac{3}{2}$. 选(A).

11. (**B**). 由 $\lim\limits_{x\to -1}f(x)=\lim\limits_{x\to -1}\frac{e^{\frac{1}{x-2}}}{x-1}\cdot\frac{\ln(-x)}{x+1}=-\frac{e^{-\frac{1}{3}}}{2}\lim\limits_{x\to -1}\frac{\ln[1-(x+1)]}{x+1}=\frac{e^{-\frac{1}{3}}}{2}$,得 $x=-1$ 为可去间断点;

由 $\lim\limits_{x\to 0}f(x)=+\infty$,得 $x=0$ 为第二类间断点;

由 $\lim\limits_{x\to 1}f(x)=\lim\limits_{x\to 1}\frac{e^{\frac{1}{x-2}}}{x+1}\cdot\frac{\ln x}{x-1}=\frac{e^{-1}}{2}\lim\limits_{x\to 1}\frac{\ln[1+(x-1)]}{x-1}=\frac{e^{-1}}{2}$,得 $x=1$ 为可去间断点;

由 $\lim\limits_{x\to 2^-}f(x)=\lim\limits_{x\to 2^-}\frac{\ln x}{x^2-1}\cdot e^{\frac{1}{x-2}}=0$,$\lim\limits_{x\to 2^+}f(x)=\lim\limits_{x\to 2^+}\frac{\ln x}{x^2-1}\cdot e^{\frac{1}{x-2}}=+\infty$,得 $x=2$ 为第二类间断点. 选(B).

12. (**B**). 当 $|x|<1$ 时,$f(x)=1+x$;当 $|x|>1$ 时,$f(x)=0$;

当 $x=-1$ 时,$f(x)=0$;当 $x=1$ 时,$f(x)=1$.

于是 $f(x)=\begin{cases}1+x, & |x|<1,\\ 0, & |x|>1,\\ 0, & x=-1,\\ 1, & x=1,\end{cases}$ 显然 $x=1$ 为函数 $f(x)$ 的间断点. 选(B).

二、填空题

1. 解 $\lim\limits_{x\to 0}\frac{e^{\tan x}-e^{\sin x}}{x^2-\ln^2(1+x)}=\lim\limits_{x\to 0}e^{\sin x}\cdot\frac{e^{\tan x-\sin x}-1}{x^2-\ln^2(1+x)}=\lim\limits_{x\to 0}\frac{\tan x-\sin x}{x^2-\ln^2(1+x)}$

$$=\lim_{x\to 0}\frac{\tan x(1-\cos x)}{x^2-\ln^2(1+x)}=\frac{1}{2}\lim_{x\to 0}\frac{x^3}{x^2-\ln^2(1+x)}$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{1}{\frac{x+\ln(1+x)}{x}\cdot\frac{x-\ln(1+x)}{x^2}}$$

$$=\frac{1}{2}\cdot\frac{1}{\lim\limits_{x\to 0}\frac{x+\ln(1+x)}{x}\cdot\lim\limits_{x\to 0}\frac{x-\ln(1+x)}{x^2}}=\frac{1}{2}\cdot\frac{1}{2\cdot\frac{1}{2}}=\frac{1}{2}.$$

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2. 解 $\lim\limits_{x\to0}\left(\dfrac{2^x+8^x}{2}\right)^{\frac{1}{x}}=\lim\limits_{x\to0}\left[\left(1+\dfrac{2^x+8^x-2}{2}\right)^{\frac{2}{2^x+8^x-2}}\right]^{\frac{1}{x}\cdot\frac{2^x+8^x-2}{2}}=\mathrm{e}^{\frac{1}{2}\lim\limits_{x\to0}\frac{2^x+8^x-2}{x}}$

$=\mathrm{e}^{\frac{1}{2}\left(\lim\limits_{x\to0}\frac{\mathrm{e}^{x\ln2}-1}{x}+\lim\limits_{x\to0}\frac{\mathrm{e}^{x\ln8}-1}{x}\right)}=\mathrm{e}^{\frac{1}{2}(\ln2+\ln8)}=4.$

3. 解 $\dfrac{2x^2+5x+3}{x-2}-ax=\dfrac{(2-a)x^2+(5+2a)x+3}{x-2}.$

由$\lim\limits_{x\to\infty}\dfrac{(2-a)x^2+(5+2a)x+3}{x-2}=b+2$，得$\begin{cases}2-a=0,\\5+2a=b+2,\end{cases}$解得$a=2,b=7$，则$ab=14$.

4. 解 由$\lim\limits_{x\to0}\dfrac{x^2}{(b-\cos x)\sqrt{a+x^2}}=1$，得$b=1$.

则$\lim\limits_{x\to0}\dfrac{x^2}{(b-\cos x)\sqrt{a+x^2}}=\dfrac{2}{\sqrt{a}}=1$，故$a=4$.

5. 解 当$x\to0$时，$(1+ax^2)^{\frac{1}{2}}-1\sim\dfrac{a}{2}x^2$，

$(1+x)^{\sin2x}-1=\mathrm{e}^{\sin2x\cdot\ln(1+x)}-1\sim\sin2x\cdot\ln(1+x)\sim2x^2$，故$a=4$.

6. 解 $\lim\limits_{x\to0}f(x)=\lim\limits_{x\to0}\dfrac{\arcsin2x^2+\mathrm{e}^{ax^2}-1}{\ln(1+2x^2)}=\lim\limits_{x\to0}\left(\dfrac{\arcsin2x^2}{2x^2}+\dfrac{\mathrm{e}^{ax^2}-1}{2x^2}\right)=1+\dfrac{a}{2}$，$f(0)=a$，

因为$f(x)$在$x=0$处连续，所以$1+\dfrac{a}{2}=a$，故$a=2$.

7. 解 $\lim\limits_{x\to0}F(x)=\lim\limits_{x\to0}\dfrac{f(x)+a\sin x}{x}=\lim\limits_{x\to0}\left[\dfrac{f(x)-f(0)}{x}+a\dfrac{\sin x}{x}\right]=f'(0)+a=a+b$，

因为$F(x)$在$x=0$处连续，所以$A=a+b$.

8. 解 令$\lim\limits_{x\to0}f(x)=A$，$f(x)=\dfrac{x\mathrm{e}^{-x^2}-\sin x}{x^3}-\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\arcsin x}\lim\limits_{x\to0}f(x)$两边取极限，得

$$A=\lim_{x\to0}\frac{x\mathrm{e}^{-x^2}-\sin x}{x^3}-A\lim_{x\to0}\frac{\sqrt{1+x}-\sqrt{1-x}}{\arcsin x}.$$

$$\lim_{x\to0}\frac{x\mathrm{e}^{-x^2}-\sin x}{x^3}=\lim_{x\to0}\frac{x\mathrm{e}^{-x^2}-x}{x^3}+\lim_{x\to0}\frac{x-\sin x}{x^3}$$

$$=\lim_{x\to0}\frac{\mathrm{e}^{-x^2}-1}{x^2}+\frac{1}{3}\lim_{x\to0}\frac{1-\cos x}{x^2}=-\frac{5}{6},$$

$\lim\limits_{x\to0}\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\arcsin x}=\lim\limits_{x\to0}\dfrac{\sqrt{1+x}-\sqrt{1-x}}{x}=2\lim\limits_{x\to0}\dfrac{1}{\sqrt{1+x}+\sqrt{1-x}}=1$，则$A=-\dfrac{5}{12}$.

三、解答题

1. 证明 (1) 对任意的$\varepsilon>0$，$\left|\dfrac{2x^2+x-3}{x-1}-5\right|=2|x-1|<\varepsilon$等价于$|x-1|<\dfrac{\varepsilon}{2}$.

取$\delta=\dfrac{\varepsilon}{2}$，则对任意的$\varepsilon>0$，存在$\delta=\dfrac{\varepsilon}{2}>0$，当$0<|x-1|<\delta$时，有$\left|\dfrac{2x^2+x-3}{x-1}-5\right|<\varepsilon$，

由极限的定义，得$\lim\limits_{x\to1}\dfrac{2x^2+x-3}{x-1}=5$.

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(2) 对任意的 $\varepsilon>0$，$\left|\dfrac{2x-1}{x+1}-2\right|=\dfrac{3}{x+1}<\dfrac{3}{x}<\varepsilon$ 等价于 $x>\dfrac{3}{\varepsilon}$，取 $X=\dfrac{3}{\varepsilon}$，则对任意的 $\varepsilon>0$，存在 $X=\dfrac{3}{\varepsilon}>0$，当 $x>X$ 时，有 $\left|\dfrac{2x-1}{x+1}-2\right|<\varepsilon$，由极限的定义，得 $\lim\limits_{x\to+\infty}\dfrac{2x-1}{x+1}=2$.

2. 解 (1) $\lim\limits_{x\to0}\dfrac{(1-x^2)^{\sin 2x}-1}{x^2\arcsin x}=\lim\limits_{x\to0}\dfrac{(1-x^2)^{\sin 2x}-1}{x^3}=\lim\limits_{x\to0}\dfrac{e^{\sin 2x\cdot\ln(1-x^2)}-1}{x^3}$

$$=\lim_{x\to0}\frac{\sin 2x\cdot\ln(1-x^2)}{x^3}=-2.$$

(2) $\lim\limits_{x\to0}\dfrac{\sqrt{1+x\cos x}-\sqrt{1+\arcsin x}}{x^3}=\lim\limits_{x\to0}\dfrac{1}{\sqrt{1+x\cos x}+\sqrt{1+\arcsin x}}\cdot\dfrac{x\cos x-\arcsin x}{x^3}$

$$=\frac{1}{2}\lim_{x\to0}\frac{x\cos x-\arcsin x}{x^3}$$

$$=\frac{1}{2}\left(\lim_{x\to0}\frac{x\cos x-x}{x^3}-\lim_{x\to0}\frac{\arcsin x-x}{x^3}\right),$$

其中 $\lim\limits_{x\to0}\dfrac{x\cos x-x}{x^3}=\lim\limits_{x\to0}\dfrac{\cos x-1}{x^2}=-\dfrac{1}{2}$，

$$\lim_{x\to0}\frac{\arcsin x-x}{x^3}=\frac{1}{3}\lim_{x\to0}\frac{(1-x^2)^{-\frac{1}{2}}-1}{x^2}=\frac{1}{3}\lim_{x\to0}\frac{-\frac{1}{2}(-x^2)}{x^2}=\frac{1}{6},$$

故原式 $=\dfrac{1}{2}\left(-\dfrac{1}{2}-\dfrac{1}{6}\right)=-\dfrac{1}{3}$.

(3) $\lim\limits_{x\to0}\dfrac{2^x+2^{-x}-2}{x^2}=\dfrac{1}{2}\lim\limits_{x\to0}\dfrac{2^x\ln 2-2^{-x}\ln 2}{x}=\dfrac{\ln 2}{2}\lim\limits_{x\to0}\dfrac{2^x-2^{-x}}{x}$

$$=\frac{\ln 2}{2}\lim_{x\to0}(2^x\ln 2+2^{-x}\ln 2)=\ln^2 2.$$

3. 解 (1) $\lim\limits_{x\to0}\dfrac{\left(\dfrac{1+\cos 2x}{2}\right)^{\sin x}-1}{x^3}=\lim\limits_{x\to0}\dfrac{e^{\sin x\cdot\ln\frac{1+\cos 2x}{2}}-1}{x^3}=\lim\limits_{x\to0}\dfrac{\sin x\cdot\ln\dfrac{1+\cos 2x}{2}}{x^3}$

$$=\lim_{x\to0}\frac{\ln\left(1+\dfrac{\cos 2x-1}{2}\right)}{x^2}=-\frac{1}{2}\lim_{x\to0}\frac{1-\cos 2x}{x^2}=-1.$$

(2) $\lim\limits_{x\to0}\dfrac{e^{\sin(\tan x)}-e^x}{x^3}=\lim e^x\cdot\lim\limits_{x\to0}\dfrac{e^{\sin(\tan x)-x}-1}{x^3}=\lim\limits_{x\to0}\dfrac{e^{\sin(\tan x)-x}-1}{x^3}=\lim\limits_{x\to0}\dfrac{\sin(\tan x)-x}{x^3}$

$$=\lim_{x\to0}\frac{\sin(\tan x)-\tan x}{x^3}+\lim_{x\to0}\frac{\tan x-x}{x^3},$$

其中 $\lim\limits_{x\to0}\dfrac{\sin(\tan x)-\tan x}{x^3}=\lim\limits_{x\to0}\dfrac{\sin(\tan x)-\tan x}{\tan^3 x}\overset{t=\tan x}{=\!=}\lim\limits_{t\to0}\dfrac{\sin t-t}{t^3}$

$$=\frac{1}{3}\lim_{t\to0}\frac{\cos t-1}{t^2}=-\frac{1}{6},$$

$$\lim_{x\to0}\frac{\tan x-x}{x^3}=\frac{1}{3}\lim_{x\to0}\frac{\sec^2 x-1}{x^2}=\frac{1}{3}\lim_{x\to0}\frac{\tan^2 x}{x^2}=\frac{1}{3},$$

故原式 $=-\dfrac{1}{6}+\dfrac{1}{3}=\dfrac{1}{6}$.

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(3) 当 $x\to1$ 时，$\sin\pi x=\sin[\pi+\pi(x-1)]=-\sin\pi(x-1)\sim-\pi(x-1)$，则

$$\lim_{x\to1}\frac{x-1-\ln x}{\sin^2\pi x}=\frac{1}{\pi^2}\lim_{x\to1}\frac{x-1-\ln x}{(x-1)^2}=\frac{1}{\pi^2}\lim_{x\to1}\frac{x-1-\ln[1+(x-1)]}{(x-1)^2}$$

$$\xlongequal{t=x-1}\frac{1}{\pi^2}\lim_{t\to0}\frac{t-\ln(1+t)}{t^2}=\frac{1}{2\pi^2}\lim_{t\to0}\frac{1-\frac{1}{1+t}}{t}=\frac{1}{2\pi^2}.$$

$$(4)\lim_{x\to0}\frac{(1+x)^{\frac{2}{x}}-e^2}{x}=\lim_{x\to0}\frac{e^{\frac{2\ln(1+x)}{x}}-e^2}{x}=e^2\lim_{x\to0}\frac{e^{\frac{2\ln(1+x)}{x}-2}-1}{x}$$

$$=e^2\lim_{x\to0}\frac{\frac{2\ln(1+x)}{x}-2}{x}=2e^2\lim_{x\to0}\frac{\ln(1+x)-x}{x^2}=-e^2.$$

4. 解 (1) **方法一** $\lim\limits_{x\to0}\dfrac{\ln(1+2x)+e^{-2x}-1}{x^2}=\lim\limits_{x\to0}\dfrac{\frac{2}{1+2x}-2e^{-2x}}{2x}=\lim\limits_{x\to0}\dfrac{\frac{1}{1+2x}-e^{-2x}}{x}$

$$=\lim_{x\to0}\frac{(1+2x)^{-1}-1}{x}-\lim_{x\to0}\frac{e^{-2x}-1}{x}$$

$$=\lim_{x\to0}\frac{-2x}{x}-\lim_{x\to0}\frac{-2x}{x}=0.$$

方法二 泰勒公式法(可在学完第三章内容后尝试用此方法).

由 $\ln(1+x)=x-\dfrac{x^2}{2}+o(x^2)$，得 $\ln(1+2x)=2x-2x^2+o(x^2)$.

再由 $e^x=1+x+\dfrac{x^2}{2!}+o(x^2)$，得 $e^{-2x}=1-2x+2x^2+o(x^2)$.

则 $\ln(1+2x)+e^{-2x}-1=o(x^2)$，故 $\lim\limits_{x\to0}\dfrac{\ln(1+2x)+e^{-2x}-1}{x^2}=0$.

$$(2)\lim_{x\to1}\frac{\sqrt{3x-2}+\sqrt{2x-1}-2}{\sin\pi x}=\lim_{x\to1}\frac{\sqrt{1+3(x-1)}+\sqrt{1+2(x-1)}-2}{\sin[\pi+\pi(x-1)]}$$

$$=-\lim_{x\to1}\frac{\sqrt{1+3(x-1)}+\sqrt{1+2(x-1)}-2}{\sin\pi(x-1)}$$

$$\xlongequal{t=x-1}-\frac{1}{\pi}\lim_{t\to0}\frac{\sqrt{1+3t}+\sqrt{1+2t}-2}{t}$$

$$=-\frac{1}{\pi}\left[\lim_{t\to0}\frac{(1+3t)^{\frac{1}{2}}-1}{t}+\lim_{t\to0}\frac{(1+2t)^{\frac{1}{2}}-1}{t}\right]$$

$$=-\frac{1}{\pi}\left[\lim_{t\to0}\frac{\frac{1}{2}\cdot3t}{t}+\lim_{t\to0}\frac{\frac{1}{2}\cdot2t}{t}\right]=-\frac{5}{2\pi}.$$

5. 解 $2=\lim\limits_{x\to0}\dfrac{xf(x)+\ln(1+2x)}{x^2}=\lim\limits_{x\to0}\dfrac{xf(x)+2x}{x^2}+\lim\limits_{x\to0}\dfrac{\ln(1+2x)-2x}{x^2}$

$$=\lim_{x\to0}\frac{f(x)+2}{x}+\lim_{x\to0}\frac{\ln(1+2x)-2x}{x^2},$$

而 $\lim\limits_{x\to0}\dfrac{\ln(1+2x)-2x}{x^2}=4\lim\limits_{x\to0}\dfrac{\ln(1+2x)-2x}{(2x)^2}\xlongequal{2x=t}4\lim\limits_{t\to0}\dfrac{\ln(1+t)-t}{t^2}=-2$，故

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$$\lim_{x\to 0}\frac{f(x)+2}{x}=4.$$

[注] 本题不可用 $\ln(1+2x)\sim 2x\,(x\to 0)$，因为 $f(x)$ 为抽象函数，精确度不确定，故在加减中不可使用等价无穷小替换.

6. 解 (1) $\lim\limits_{x\to\infty}\left(\cos\frac{1}{x}\right)^{x^2}=\lim\limits_{x\to\infty}\left\{\left[1+\left(\cos\frac{1}{x}-1\right)\right]^{\frac{1}{\cos\frac{1}{x}-1}}\right\}^{x^2\left(\cos\frac{1}{x}-1\right)}$

$$=\mathrm{e}^{\lim\limits_{x\to\infty}x^2\left(\cos\frac{1}{x}-1\right)}=\mathrm{e}^{\lim\limits_{x\to\infty}\frac{\cos\frac{1}{x}-1}{\frac{1}{x^2}}}\xlongequal{t=\frac{1}{x}}\mathrm{e}^{\lim\limits_{t\to 0}\frac{\cos t-1}{t^2}}=\mathrm{e}^{-\frac{1}{2}}.$$

(2) $\lim\limits_{x\to 0}(\cos 2x+\mathrm{e}^{x^2}-1)^{\frac{1}{x-\ln(1+x)}}=\lim\limits_{x\to 0}\left\{[1+(\cos 2x+\mathrm{e}^{x^2}-2)]^{\frac{1}{\cos 2x+\mathrm{e}^{x^2}-2}}\right\}^{\frac{\cos 2x+\mathrm{e}^{x^2}-2}{x-\ln(1+x)}}$

$$=\mathrm{e}^{\lim\limits_{x\to 0}\frac{\cos 2x+\mathrm{e}^{x^2}-2}{x-\ln(1+x)}}=\mathrm{e}^{\lim\limits_{x\to 0}\frac{\cos 2x+\mathrm{e}^{x^2}-2}{\frac{1}{2}x^2}}=\mathrm{e}^{2\left(\lim\limits_{x\to 0}\frac{\cos 2x-1}{x^2}+\lim\limits_{x\to 0}\frac{\mathrm{e}^{x^2}-1}{x^2}\right)}$$

$$=\mathrm{e}^{2\left[\lim\limits_{x\to 0}\frac{-\frac{1}{2}(2x)^2}{x^2}+1\right]}=\mathrm{e}^{-2}.$$

(3) $\lim\limits_{x\to 0}\left(\frac{1+\arctan x}{1+\sin x}\right)^{\frac{1}{x^3}}=\lim\limits_{x\to 0}\left[\left(1+\frac{\arctan x-\sin x}{1+\sin x}\right)^{\frac{1+\sin x}{\arctan x-\sin x}}\right]^{\frac{1}{x^3}\cdot\frac{\arctan x-\sin x}{1+\sin x}}$

$$=\mathrm{e}^{\lim\limits_{x\to 0}\frac{1}{1+\sin x}\cdot\frac{\arctan x-\sin x}{x^3}}=\mathrm{e}^{\lim\limits_{x\to 0}\frac{\arctan x-\sin x}{x^3}},$$

$$\lim_{x\to 0}\frac{\arctan x-\sin x}{x^3}=\frac{1}{3}\lim_{x\to 0}\frac{\frac{1}{1+x^2}-\cos x}{x^2}=\frac{1}{3}\left[\lim_{x\to 0}\frac{(1+x^2)^{-1}-1}{x^2}+\lim_{x\to 0}\frac{1-\cos x}{x^2}\right]$$

$$=\frac{1}{3}\left(-1+\frac{1}{2}\right)=-\frac{1}{6},$$

故 $\lim\limits_{x\to 0}\left(\frac{1+\arctan x}{1+\sin x}\right)^{\frac{1}{x^3}}=\mathrm{e}^{-\frac{1}{6}}$.

(4) $\lim\limits_{x\to 0}\left(\frac{a_1^x+a_2^x+\cdots+a_n^x}{n}\right)^{\frac{1}{x}}=\lim\limits_{x\to 0}\left[\left(1+\frac{a_1^x+a_2^x+\cdots+a_n^x-n}{n}\right)^{\frac{n}{a_1^x+a_2^x+\cdots+a_n^x-n}}\right]^{\frac{a_1^x+a_2^x+\cdots+a_n^x-n}{nx}}$

$$=\mathrm{e}^{\frac{1}{n}\lim\limits_{x\to 0}\frac{a_1^x+a_2^x+\cdots+a_n^x-n}{x}}=\mathrm{e}^{\frac{1}{n}\left(\lim\limits_{x\to 0}\frac{\mathrm{e}^{x\ln a_1}-1}{x}+\lim\limits_{x\to 0}\frac{\mathrm{e}^{x\ln a_2}-1}{x}+\cdots+\lim\limits_{x\to 0}\frac{\mathrm{e}^{x\ln a_n}-1}{x}\right)}$$

$$=\mathrm{e}^{\frac{1}{n}(\ln a_1+\ln a_2+\cdots+\ln a_n)}=\sqrt[n]{a_1a_2\cdots a_n}.$$

(5) $\lim\limits_{x\to 1}\left(\tan\frac{\pi}{4}x\right)^{\tan\frac{\pi}{2}x}=\lim\limits_{x\to 1}\left\{\left[1+\left(\tan\frac{\pi}{4}x-1\right)\right]^{\frac{1}{\tan\frac{\pi}{4}x-1}}\right\}^{\tan\frac{\pi}{2}x\cdot\left(\tan\frac{\pi}{4}x-1\right)}=\mathrm{e}^{\lim\limits_{x\to 1}\tan\frac{\pi}{2}x\cdot\left(\tan\frac{\pi}{4}x-1\right)}$,

由 $\lim\limits_{x\to 1}\tan\frac{\pi}{2}x\cdot\left(\tan\frac{\pi}{4}x-1\right)=\lim\limits_{x\to 1}\sin\frac{\pi}{2}x\cdot\frac{\tan\frac{\pi}{4}x-1}{\cos\frac{\pi}{2}x}=\lim\limits_{x\to 1}\frac{\frac{\pi}{4}\sec^2\frac{\pi}{4}x}{-\frac{\pi}{2}\sin\frac{\pi}{2}x}=-1$，得

$$\lim_{x\to 1}\left(\tan\frac{\pi}{4}x\right)^{\tan\frac{\pi}{2}x}=\mathrm{e}^{-1}.$$

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7. 解 (1) 由$(x+2)^4=C_4^0x^4+C_4^1\cdot 2x^3+\cdots=x^4+8x^3+\cdots$,

$(x-1)^4=C_4^0x^4-C_4^1x^3+\cdots=x^4-4x^3+\cdots$,得$(x+2)^4-(x-1)^4=12x^3+\cdots$.

则$\lim\limits_{x\to\infty}\dfrac{(2x^2+3x+2)(4x+3)}{(x+2)^4-(x-1)^4}=\lim\limits_{x\to\infty}\dfrac{8x^3+\cdots}{12x^3+\cdots}=\dfrac{2}{3}$.

(2) $\lim\limits_{x\to+\infty}\dfrac{\ln(2x^3+3x^2+5)}{\ln(5x^6-x+3)}=\lim\limits_{x\to+\infty}\dfrac{\dfrac{6x^2+6x}{2x^3+3x^2+5}}{\dfrac{30x^5-1}{5x^6-x+3}}=\lim\limits_{x\to+\infty}\dfrac{30x^8+\cdots}{60x^8+\cdots}=\dfrac{1}{2}$.

(3) 当$x\to+\infty$时,$e^{x^2}\leqslant e^{x^2}+x\leqslant 2e^{x^2}$,则$x^2\leqslant\ln(e^{x^2}+x)\leqslant x^2+\ln 2$,故$1\leqslant\dfrac{\ln(e^{x^2}+x)}{x^2}\leqslant 1+\dfrac{\ln 2}{x^2}$,由夹逼准则,得$\lim\limits_{x\to+\infty}\dfrac{\ln(e^{x^2}+x)}{x^2}=1$,则

$$\lim_{x\to+\infty}\frac{3x^2+2x\sin x+1}{\ln(e^{x^2}+x)}=\lim_{x\to+\infty}\frac{\dfrac{3x^2+2x\sin x+1}{x^2}}{\dfrac{\ln(e^{x^2}+x)}{x^2}}=\lim_{x\to+\infty}\frac{3x^2+2x\sin x+1}{x^2}$$

$$=\lim_{x\to+\infty}\left(3+\frac{2}{x}\sin x+\frac{1}{x^2}\right)=3.$$

8. 解 $\lim\limits_{x\to+\infty}\dfrac{\left(1+\frac{1}{x}\right)^{x^2}}{e^x}=\lim\limits_{x\to+\infty}\dfrac{e^{x^2\ln\left(1+\frac{1}{x}\right)}}{e^x}=\lim\limits_{x\to+\infty}e^{x^2\ln\left(1+\frac{1}{x}\right)-x}=e^{\lim\limits_{x\to+\infty}x^2\left[\ln\left(1+\frac{1}{x}\right)-\frac{1}{x}\right]}$

$$=e^{\lim\limits_{x\to+\infty}\frac{\ln\left(1+\frac{1}{x}\right)-\frac{1}{x}}{\frac{1}{x^2}}}\xlongequal{t=\frac{1}{x}}e^{\lim\limits_{t\to0^+}\frac{\ln(1+t)-t}{t^2}}=e^{\frac{1}{2}\lim\limits_{t\to0^+}\frac{\frac{1}{1+t}-1}{t}}=e^{-\frac{1}{2}}.$$

[注]本题不能用$\left[\left(1+\dfrac{1}{x}\right)^x\right]^x\to e^x(x\to+\infty)$,因为复合函数$f(x)^{g(x)}$极限计算中要求$\lim f(x)$,$\lim g(x)$均存在,本题中$x\to+\infty$.

9. 解 (1) $\lim\limits_{x\to\infty}x^2\left[\dfrac{1}{x}-\ln\left(1+\dfrac{1}{x}\right)\right]=\lim\limits_{x\to\infty}\dfrac{\dfrac{1}{x}-\ln\left(1+\dfrac{1}{x}\right)}{\dfrac{1}{x^2}}\xlongequal{t=\frac{1}{x}}\lim\limits_{t\to0}\dfrac{t-\ln(1+t)}{t^2}$

$$=\lim_{t\to0}\frac{1-\dfrac{1}{1+t}}{2t}=\frac{1}{2}.$$

(2) 令$f(t)=\sin t$,则$f'(t)=\cos t$,由拉格朗日中值定理,得

$$\sin\frac{2}{x-1}-\sin\frac{1}{2x+1}=f\left(\frac{2}{x-1}\right)-f\left(\frac{1}{2x+1}\right)=f'(\xi)\left(\frac{2}{x-1}-\frac{1}{2x+1}\right)$$

$$=\frac{3x+3}{(x-1)(2x+1)}\cos\xi,$$

其中ξ介于$\dfrac{1}{2x+1}$与$\dfrac{2}{x-1}$之间,当$x\to+\infty$时,$\xi\to0$,则

$$\lim_{x\to+\infty}x\left(\sin\frac{2}{x-1}-\sin\frac{1}{2x+1}\right)=\lim_{x\to+\infty}\frac{x(3x+3)}{(x-1)(2x+1)}\cos\xi=\frac{3}{2}.$$

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10. 解 (1) **方法一**
$$\lim_{x\to0}\left(\frac{1}{\sin^2x}-\frac{\cos^2x}{x^2}\right)=\lim_{x\to0}\frac{x^2-\sin^2x\cos^2x}{x^2\sin^2x}=\lim_{x\to0}\frac{x^2-\sin^2x\cos^2x}{x^4}$$
$$=\lim_{x\to0}\left(1+\frac{\sin x}{x}\cdot\cos x\right)\cdot\frac{x-\sin x\cos x}{x^3}$$
$$=2\lim_{x\to0}\frac{x-\sin x\cos x}{x^3}$$
$$=2\left(\lim_{x\to0}\frac{x-\sin x}{x^3}+\lim_{x\to0}\frac{\sin x-\sin x\cos x}{x^3}\right)$$
$$=2\left(\frac{1}{6}+\lim_{x\to0}\frac{\sin x}{x}\cdot\frac{1-\cos x}{x^2}\right)$$
$$=2\left(\frac{1}{6}+\frac{1}{2}\right)=\frac{4}{3}.$$

方法二
$$\lim_{x\to0}\left(\frac{1}{\sin^2x}-\frac{\cos^2x}{x^2}\right)=\lim_{x\to0}\frac{x^2-\sin^2x\cos^2x}{x^2\sin^2x}=\lim_{x\to0}\frac{x^2-\frac{1}{4}\sin^2 2x}{x^4}$$
$$=4\lim_{x\to0}\frac{(2x)^2-\sin^2 2x}{(2x)^4}\xlongequal{t=2x}4\lim_{t\to0}\frac{t^2-\sin^2t}{t^4}$$
$$=4\lim_{t\to0}\frac{t+\sin t}{t}\cdot\frac{t-\sin t}{t^3}=\frac{4}{3}.$$

(2) $$\lim_{x\to0}\left(\frac{1}{x^2}-\frac{1}{\arcsin^2x}\right)=\lim_{x\to0}\frac{\arcsin^2x-x^2}{x^2\arcsin^2x}=\lim_{x\to0}\frac{\arcsin^2x-x^2}{x^4}$$
$$=\lim_{x\to0}\frac{\arcsin x+x}{x}\cdot\frac{\arcsin x-x}{x^3}=2\lim_{x\to0}\frac{\arcsin x-x}{x^3}$$
$$=\frac{2}{3}\lim_{x\to0}\frac{\frac{1}{\sqrt{1-x^2}}-1}{x^2}=-\frac{2}{3}\lim_{x\to0}\frac{\sqrt{1-x^2}-1}{x^2\sqrt{1-x^2}}$$
$$=-\frac{2}{3}\lim_{x\to0}\frac{\sqrt{1-x^2}-1}{x^2}=-\frac{2}{3}\lim_{x\to0}\frac{-\frac{1}{2}x^2}{x^2}=\frac{1}{3}.$$

(3) **方法一**
$$\lim_{x\to-\infty}(\sqrt{4x^2-4x+1}-\sqrt{4x^2+2x})$$
$$=\lim_{x\to-\infty}\frac{-6x+1}{\sqrt{4x^2-4x+1}+\sqrt{4x^2+2x}}$$
$$=\lim_{x\to-\infty}\frac{-6+\frac{1}{x}}{-\sqrt{4-\frac{4}{x}+\frac{1}{x^2}}-\sqrt{4+\frac{2}{x}}}=\frac{3}{2}.$$

方法二
$$\lim_{x\to-\infty}(\sqrt{4x^2-4x+1}-\sqrt{4x^2+2x})$$
$$=\lim_{x\to-\infty}2x\left(\sqrt{1+\frac{1}{2x}}-\sqrt{1-\frac{1}{x}+\frac{1}{4x^2}}\right)$$
$$\xlongequal{t=\frac{1}{x}}2\lim_{t\to0^-}\frac{\sqrt{1+\frac{t}{2}}-\sqrt{1-t+\frac{t^2}{4}}}{t}$$

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$$=2\lim_{t\to0^-}\frac{\left(1+\frac{t}{2}\right)^{\frac{1}{2}}-1-\left[\left(1-t+\frac{t^2}{4}\right)^{\frac{1}{2}}-1\right]}{t}=\frac{3}{2}.$$

(4) $\lim\limits_{x\to+\infty}(2x\arctan x-\pi x)=2\lim\limits_{x\to+\infty}x\left(\arctan x-\frac{\pi}{2}\right)$

$$=2\lim_{x\to+\infty}\frac{\arctan x-\frac{\pi}{2}}{\frac{1}{x}}=2\lim_{x\to+\infty}\frac{\frac{1}{1+x^2}}{-\frac{1}{x^2}}$$

$$=-2\lim_{x\to+\infty}\frac{x^2}{1+x^2}=-2.$$

11. 解 $\lim\limits_{x\to0^+}x^{\sin\frac{2x}{1+x^2}}=e^{\lim\limits_{x\to0^+}\sin\frac{2x}{1+x^2}\cdot\ln x}=e^{\lim\limits_{x\to0^+}\frac{2x}{1+x^2}\cdot\ln x}=e^{2\lim\limits_{x\to0^+}x\cdot\ln x}$

$$=e^{2\lim\limits_{x\to0^+}\frac{\ln x}{\frac{1}{x}}}=e^{2\lim\limits_{x\to0^+}\frac{\frac{1}{x}}{-\frac{1}{x^2}}}=e^{-2\lim\limits_{x\to0^+}x}=e^0=1.$$

12. 解 $\lim\limits_{x\to0^+}\left(\frac{1}{\tan x}\right)^{\sqrt{x}}=e^{\lim\limits_{x\to0^+}\sqrt{x}\ln\frac{1}{\tan x}}=e^{-\lim\limits_{x\to0^+}\sqrt{x}\ln(\tan x)}=e^{-\lim\limits_{x\to0^+}\frac{\ln(\tan x)}{\frac{1}{\sqrt{x}}}}$

$$=e^{-\lim\limits_{x\to0^+}\frac{\frac{\sec^2x}{\tan x}}{-\frac{1}{2x\sqrt{x}}}}=e^{2\lim\limits_{x\to0^+}\frac{x\sqrt{x}}{\tan x}}=e^{2\lim\limits_{x\to0^+}\sqrt{x}}=e^0=1.$$

13. 解 (1) $\frac{1}{1\times2}+\frac{1}{2\times3}+\cdots+\frac{1}{n(n+1)}=\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{n}-\frac{1}{n+1}\right)=1-\frac{1}{n+1}$,则

$$\lim_{n\to\infty}\left[\frac{1}{1\times2}+\frac{1}{2\times3}+\cdots+\frac{1}{n(n+1)}\right]^{\frac{2n^2}{n-2}}=\lim_{n\to\infty}\left[\left(1-\frac{1}{n+1}\right)^{-(n+1)}\right]^{-\frac{2n^2}{(n+1)(n-2)}}=e^{-2}.$$

(2) $\frac{1}{n^2+n+1}+\frac{2}{n^2+n+2}+\cdots+\frac{n}{n^2+n+n}\leqslant\frac{1+2+\cdots+n}{n^2+n+1}=\frac{\frac{1}{2}n(n+1)}{n^2+n+1}$,

$$\frac{1}{n^2+n+1}+\frac{2}{n^2+n+2}+\cdots+\frac{n}{n^2+n+n}\geqslant\frac{1+2+\cdots+n}{n^2+n+n}=\frac{\frac{1}{2}n(n+1)}{n^2+n+n},$$

$$\frac{\frac{1}{2}n(n+1)}{n^2+n+n}\leqslant\frac{1}{n^2+n+1}+\frac{2}{n^2+n+2}+\cdots+\frac{n}{n^2+n+n}\leqslant\frac{\frac{1}{2}n(n+1)}{n^2+n+1},$$

因为$\lim\limits_{n\to\infty}\frac{\frac{1}{2}n(n+1)}{n^2+n+n}=\lim\limits_{n\to\infty}\frac{\frac{1}{2}n(n+1)}{n^2+n+1}=\frac{1}{2}$,所以由夹逼准则,得原式$=\frac{1}{2}$.

(3) 原式$=\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\frac{i}{n^2+i^2}=\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{i=1}^{n}\frac{\frac{i}{n}}{1+\left(\frac{i}{n}\right)^2}=\int_0^1\frac{x}{1+x^2}\mathrm{d}x=\frac{1}{2}\ln(1+x^2)\Big|_0^1$

$$=\frac{1}{2}\ln2.$$

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(4) 原式$=\lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{i=1}^{n}\dfrac{i}{n}\sin^2\dfrac{\pi i}{n}=\int_0^1 x\sin^2\pi x\,\mathrm{d}x=\dfrac{1}{\pi^2}\int_0^1\pi x\sin^2\pi x\,\mathrm{d}(\pi x)$

$=\dfrac{1}{\pi^2}\int_0^{\pi}x\sin^2 x\,\mathrm{d}x=\dfrac{1}{\pi^2}\cdot\dfrac{\pi}{2}\int_0^{\pi}\sin^2 x\,\mathrm{d}x=\dfrac{1}{2\pi}\cdot 2\int_0^{\frac{\pi}{2}}\sin^2 x\,\mathrm{d}x=\dfrac{1}{\pi}\cdot\dfrac{1}{2}\cdot\dfrac{\pi}{2}=\dfrac{1}{4}.$

(5)$\lim\limits_{n\to\infty}\sqrt[n]{\left(1+\dfrac{1^2}{n^2}\right)\left(1+\dfrac{2^2}{n^2}\right)\cdots\left(1+\dfrac{n^2}{n^2}\right)}=\mathrm{e}^{\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{i=1}^{n}\ln\left(1+\frac{i^2}{n^2}\right)}=\mathrm{e}^{\int_0^1\ln(1+x^2)\,\mathrm{d}x}$. 由

$\int_0^1\ln(1+x^2)\,\mathrm{d}x=x\ln(1+x^2)\Big|_0^1-\int_0^1 x\cdot\dfrac{2x}{1+x^2}\mathrm{d}x=\ln 2-2\int_0^1\left(1-\dfrac{1}{1+x^2}\right)\mathrm{d}x$

$=\ln 2-2+\dfrac{\pi}{2},$

得$\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\sqrt[n]{\left(1+\dfrac{1^2}{n^2}\right)\left(1+\dfrac{2^2}{n^2}\right)\cdots\left(1+\dfrac{n^2}{n^2}\right)}=\mathrm{e}^{\ln 2-2+\frac{\pi}{2}}$.

14. 解

$$\sum_{i=1}^{2n}\frac{i}{n^2+1}\left(1+\cos\frac{\pi i}{n}\right)^2\leqslant\sum_{i=1}^{2n}\frac{i}{n^2}\left(1+\cos\frac{\pi i}{n}\right)^2,$$

$$\sum_{i=1}^{2n}\frac{i}{n^2+1}\left(1+\cos\frac{\pi i}{n}\right)^2\geqslant\frac{n^2}{n^2+n}\cdot\sum_{i=1}^{2n}\frac{i}{n^2}\left(1+\cos\frac{\pi i}{n}\right)^2,$$

由$\lim\limits_{n\to\infty}\dfrac{n^2}{n^2+n}=1$,得

原式$=\lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{i=1}^{2n}\dfrac{i}{n}\left(1+\cos\dfrac{\pi i}{n}\right)^2=\int_0^2 x(1+\cos\pi x)^2\,\mathrm{d}x$

$=\dfrac{1}{\pi^2}\int_0^2\pi x(1+\cos\pi x)^2\,\mathrm{d}(\pi x)=\dfrac{1}{\pi^2}\int_0^{2\pi}x(1+\cos x)^2\,\mathrm{d}x$

$\xlongequal{x-\pi=t}\dfrac{1}{\pi^2}\int_{-\pi}^{\pi}(\pi+t)(1-\cos t)^2\,\mathrm{d}t$

$=\dfrac{1}{\pi}\int_{-\pi}^{\pi}(1-\cos t)^2\,\mathrm{d}t=\dfrac{2}{\pi}\int_0^{\pi}(1-\cos t)^2\,\mathrm{d}t=\dfrac{4}{\pi}\int_0^{\pi}\left(2\sin^2\dfrac{t}{2}\right)^2\mathrm{d}\left(\dfrac{t}{2}\right)$

$=\dfrac{16}{\pi}\int_0^{\frac{\pi}{2}}\sin^4 t\,\mathrm{d}t=\dfrac{16}{\pi}\cdot\dfrac{3}{4}\cdot\dfrac{1}{2}\cdot\dfrac{\pi}{2}=3.$

15. 证明 $a_{n+1}=\sqrt{2+a_n}$,显然数列$\{a_n\}$单调增加.

因为$a_1=\sqrt{2}\leqslant 2$,设$a_k\leqslant 2$,则$a_{k+1}=\sqrt{2+a_k}\leqslant\sqrt{2+2}=2$,由数学归纳法,对一切的$n$,有$a_n\leqslant 2$,即数列$\{a_n\}$单调增加有上界,故$\lim\limits_{n\to\infty}a_n$存在.

令$\lim\limits_{n\to\infty}a_n=A$,$a_{n+1}=\sqrt{2+a_n}$两边同时取极限,得$A=\sqrt{2+A}$,解得$A=2$.

16. 证明 显然$a_n>0$,$a_{n+1}=\sqrt{a_n(2-a_n)}\leqslant\dfrac{a_n+(2-a_n)}{2}=1$.

又由$a_{n+1}-a_n=\sqrt{a_n(2-a_n)}-a_n=\dfrac{2a_n(1-a_n)}{\sqrt{a_n(2-a_n)}+a_n}\geqslant 0$,得数列$\{a_n\}$单调增加.

根据单调有界定理知$\lim\limits_{n\to\infty}a_n$存在.

17. 证明 $a_1>0$,设$a_k>0$,则$a_{k+1}=\ln(1+a_k)>0$,由数学归纳法,得$a_n>0$.

因为当$x>0$时,$\ln(1+x)<x$,所以$a_{n+1}=\ln(1+a_n)<a_n$,即$\{a_n\}$单调减少,故$\lim\limits_{n\to\infty}a_n$存在.

令$\lim\limits_{n\to\infty}a_n=A$,对$a_{n+1}=\ln(1+a_n)$两边同时求极限得$A=\ln(1+A)$,解得$A=0$,即$\lim\limits_{n\to\infty}a_n=0$.

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18. 证明 显然 $a_n>0$，由 $a_n+\dfrac{1}{a_n}\geqslant 2$，得 $a_{n+1}\geqslant 1$.

由 $a_{n+1}-a_n=\dfrac{1}{2}\left(a_n+\dfrac{1}{a_n}\right)-a_n=\dfrac{1-a_n^2}{2a_n}\leqslant 0$，得数列 $\{a_n\}$ 单调减少，即数列 $\{a_n\}$ 单调减少有下界，故 $\lim\limits_{n\to\infty}a_n$ 存在.

19. 证明 因为 $f(x)\in C[0,2]$，所以 $f(x)$ 在 $[0,2]$ 上存在最小值 m 和最大值 M，显然

$$6m\leqslant f(0)+2f(1)+3f(2)\leqslant 6M,$$

即 $m\leqslant 1\leqslant M$，由介值定理，存在 $c\in[0,2]$，使得 $f(c)=1$.

20. 解 因为 $f(x)$ 为初等函数，所以 $f(x)$ 的间断点为 $x=0$ 和 $x=1$.

因为当 $x\to 0$ 时，$1-\mathrm{e}^{\frac{x}{1-x}}\sim-\dfrac{x}{1-x}\sim -x$，所以 $\lim\limits_{x\to 0}f(x)=-1$，即 $x=0$ 为 $f(x)$ 的可去间断点.

因为 $f(1-0)=\lim\limits_{x\to 1^-}\dfrac{x}{1-\mathrm{e}^{\frac{x}{1-x}}}=0$，$f(1+0)=\lim\limits_{x\to 1^+}\dfrac{x}{1-\mathrm{e}^{\frac{x}{1-x}}}=1$，所以 $x=1$ 为 $f(x)$ 的跳跃间断点.

21. 解 显然 $x=0$，$x=1$ 为函数 $f(x)$ 的间断点.

$$f(0-0)=\lim_{x\to 0^-}f(x)=-\frac{\pi}{2}\cdot\frac{1-\mathrm{e}^{-1}}{1+\mathrm{e}^{-2}},\quad f(0+0)=\lim_{x\to 0^+}f(x)=\frac{\pi}{2}\cdot\frac{1-\mathrm{e}^{-1}}{1+\mathrm{e}^{-2}},$$

因为 $f(0-0)\neq f(0+0)$，所以 $x=0$ 为 $f(x)$ 的跳跃间断点.

$$f(1-0)=\lim_{x\to 1^-}f(x)=\frac{\pi}{4},\quad f(1+0)=\lim_{x\to 1^+}f(x)=\frac{\pi}{4}\lim_{x\to 1^+}\frac{\mathrm{e}^{-\frac{1}{x-1}}-1}{\mathrm{e}^{-\frac{1}{x-1}}+\mathrm{e}^{\frac{1}{x-1}}}=0,$$

因为 $f(1-0)\neq f(1+0)$，所以 $x=1$ 为 $f(x)$ 的跳跃间断点.

22. 解 $x=0$，$x=1$ 为 $f(x)$ 的间断点.

$$\lim_{x\to 0}f(x)=\lim_{x\to 0}\frac{\ln|x|}{|x-1|}\sin x=\lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{\ln|x|}{\frac{1}{x}}=\lim_{x\to 0}\frac{\frac{1}{x}}{-\frac{1}{x^2}}=0,$$

则 $x=0$ 为 $f(x)$ 的可去间断点.

$\lim\limits_{x\to 1^-}f(x)=\lim\limits_{x\to 1^-}\dfrac{\ln x}{1-x}\sin x=\sin 1\cdot\lim\limits_{x\to 1^-}\dfrac{\ln[1+(x-1)]}{1-x}=-\sin 1$，即 $f(1-0)=-\sin 1$，

$\lim\limits_{x\to 1^+}f(x)=\lim\limits_{x\to 1^+}\dfrac{\ln x}{x-1}\sin x=\sin 1\cdot\lim\limits_{x\to 1^+}\dfrac{\ln[1+(x-1)]}{x-1}=\sin 1$，即 $f(1+0)=\sin 1$，

因为 $f(1-0)\neq f(1+0)$，所以 $x=1$ 为 $f(x)$ 的跳跃间断点.

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